

Understudy-CAD: post-training a 4B vision model to read engineering drawings into CAD programs

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Abstract

Frontier models already turn a written CAD specification into code almost perfectly: Kimi K3 gets 95% of our held-out parts right from text. From a *drawing* of the same part they do much worse. On 497 held-out parts rendered as a four-view drawing sheet, GLM-5.3 Flash gets 66.2% and Kimi K3 61.8%, and both fall to 19–37% once a part has more than six faces. We post-trained Qwen3-VL-4B-Instruct on 30,260 rendered sheets paired with their reference CadQuery programs (LoRA, one epoch, 2.9 hours on one H100) and evaluated it on exactly the same parts with an executable metric: a program counts as correct only if it runs and the solid it builds overlaps the reference by at least 90% (volumetric IoU, aligned over the 24 axis rotations). The model reaches 70.4% with a single sample at \$0.19 per 1,000 parts, and 79.7% when it samples eight programs and picks the one whose rendering best matches the input drawing — a check that needs no answer key. That is 13.5 points above the best frontier model at comparable cost, and 5–194× cheaper per part. The margin is unchanged on the subset of parts with no near-duplicate in training.

1 The task

The input is a drawing sheet: front, top and right orthographic views at one shared scale plus an isometric view, and the part’s bounding box as text. The output is a CadQuery (Python) program that builds the part. Figure 1 shows one held-out example with our model’s answer.

We chose this task after piloting the obvious one. With an explicit written specification, frontier models are at 87.5–95%, leaving no room for a smaller model to win. Replacing the text with a drawing of the same part drops them to 61–66%, and to 19–37% on parts with more than six faces. Reading a drawing is also the job an engineer actually has.

2 Data

The source is CAD-Coder (Apache-2.0, derived from Text2CAD/DeepCAD): CAD parts with a written specification and a reference CadQuery program.

Audit. We executed all 82,659 reference programs and compared each resulting solid against the size its own specification states. In the official `test` split, **14% of the checkable references contradict their own specification**; in the curated `train_high` split, 2.3%. Some references ignore the specification’s rotations, and some write files to disk. The benchmark is therefore drawn from `train_high`, and the sandbox strips export calls (Section 4).

*Code, data-building scripts and every benchmark run: <https://github.com/Damadimo/understudy>

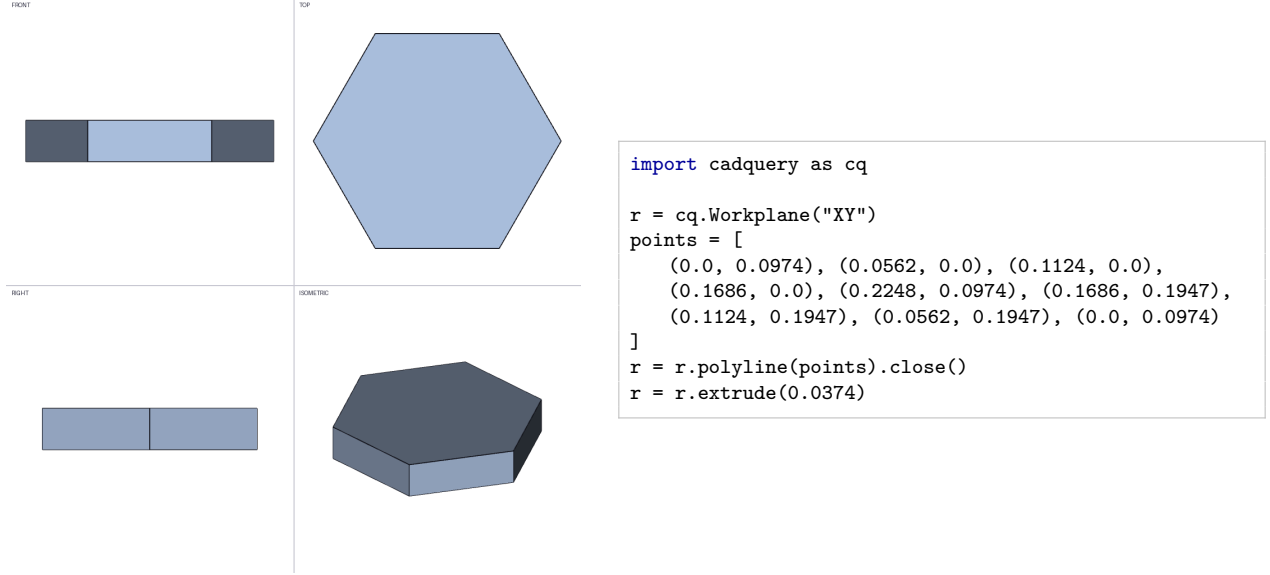


Figure 1: A held-out part (`train_high:03708`, 8 faces) and the program our model wrote for it. The prompt adds one line of text: `Bounding box: X 0.2248, Y 0.1947, Z 0.0374`. The reference program lists six hexagon corners and then translates the solid; ours lists nine points and skips the translation. The text differs, the solid does not (IoU 1.000), which is why the metric executes code instead of comparing strings.

Drawings sidestep the bad text. Every sheet is rendered from the part’s own reference program, so the picture and the answer always agree even where the dataset’s prose does not. Rendering uses one orthographic scale across the three views, which is what makes elongated parts hard: the profile of a long bar occupies a few dozen pixels.

Held-out rules. The 500 benchmark parts are grouped by source part, so no benchmark part’s source appears in training (0 shared). Every training part whose geometry signature (rounded bounding-box extents and face count) matches a benchmark part was dropped: 1,886 parts. Table 1 lists what remains.

	Parts	Notes
Audited reference programs	82,659	<code>train_high</code> 8,177, <code>train_middle</code> 66,532, <code>test</code> 7,950
Benchmark (held out)	500	322 simple (≤ 6 faces), 119 medium (7–12), 59 complex (≥ 13), 43 multi-part
Training, <code>train_high</code>	5,698	remainder after removing benchmark sources and geometries
Training, <code>train_middle</code> #1	12,000	70% medium/complex/multi-part
Training, <code>train_middle</code> #2	12,868	every remaining hard part, one per geometry signature
Training sheets used	30,260	plus 306 for validation; 72% medium/complex, 35% multi-part

Table 1: Data inventory. Training rows are (sheet, bounding box) $\rightarrow$ reference program.

3 Model and training

We fine-tune `Qwen3-VL-4B-Instruct` with LoRA (rank 64, $\alpha = 32$, dropout 0.05) on the language model’s projections only; the vision encoder and merger stay frozen. The loss covers the program text, never the prompt or the image tokens. Each sheet enters at 1024×1024 , which the processor

turns into exactly 1,024 visual tokens; rows are 1,569 tokens at the median. One epoch over 30,260 rows is 1,892 steps at an effective batch of 16, learning rate 2×10^{-4} with a cosine schedule, and takes 2.9 hours on one H100 (Baseten Training). The run ends by merging the adapter into the base weights, which vLLM then serves directly, and by grading itself on the 500 held-out sheets inside the same job.

4 Grading

A candidate program is executed in a sandbox: an import allowlist (`cadquery`, `math`, `numpy`), no dunder access, no file I/O, export calls stripped, each task in a separate process with a timeout, run in a throwaway directory. The resulting solid is compared with the reference by exact OpenCascade boolean volumetric IoU, maximised over the 24 axis-aligned rotations after centring, because the dataset applies its own global transforms inconsistently. **A part counts as correct when the code runs and that aligned IoU is ≥ 0.9** ; a part 10% too large scores 0.75 and fails. We also record IoU ≥ 0.95 , run rate, mean IoU, Chamfer distance, latency and cost.

Rows where an API returned no answer at all (HTTP 402/5xx, dropped connections after retries) are excluded from every score and reported separately, rather than counted as model failures. This mattered: a credit limit silently turned 32 frontier calls into 402 errors mid-benchmark.

5 Test-time verification

A 4B model is cheap enough to answer a part several times. To choose among candidates without an answer key, we build each one and render it exactly as the input sheet was rendered, then score the match: the mean silhouette IoU across the three orthographic views, multiplied by how well the candidate’s bounding box matches the one given in the prompt. On 689 real frontier answers this separates correct from incorrect with AUC 0.959, and where several answers exist for one part it picks a correct one 98.1% of the time (random choice: 76.3%). Adding visible-edge matching was measured and made it worse (AUC 0.81 alone), since the same solid built a different way has different internal seams.

6 Results

Table 2 and Figure 2 give the main comparison: every lane on the same 497 held-out parts, every frontier lane at high reasoning effort and with two worked examples, ours with none.

	All [95% CI]	Simple	Medium	Complex	Multi	Runs	\$/1k	p50 / p95
Ours, best of 8	79.7 [76–83]	96.6	58.8	30.5	53.5	100.0	1.23	2.2 / 22.1 s
Ours, 1 sample	70.4 [66–74]	89.3	44.5	20.3	46.5	98.4	0.19	1.9 / 5.3 s
GLM-5.3 Flash (high)	66.2 [62–70]	85.9	37.0	18.6	39.5	98.6	1.00	3.7 / 46.2 s
Kimi K3 (high)	61.8 [58–66]	81.5	30.3	18.6	39.5	95.8	36.82	10.4 / 81.9 s
Qwen3-VL-4B, untuned	14.7 [11–18]	19.4	8.4	1.7	16.3	34.2	—	20.3 / 157.5 s

Table 2: Correct parts (%) on 497 held-out parts: code runs and aligned IoU ≥ 0.9 . Tiers by reference face count (319 simple, 119 medium, 59 complex, 43 multi-part). “Runs” is the share of programs that build a solid. Cost is API list price for the frontier lanes and amortised GPU price for ours (one H100, \$6.50/h, measured throughput). Confidence intervals are bootstrap 95%.

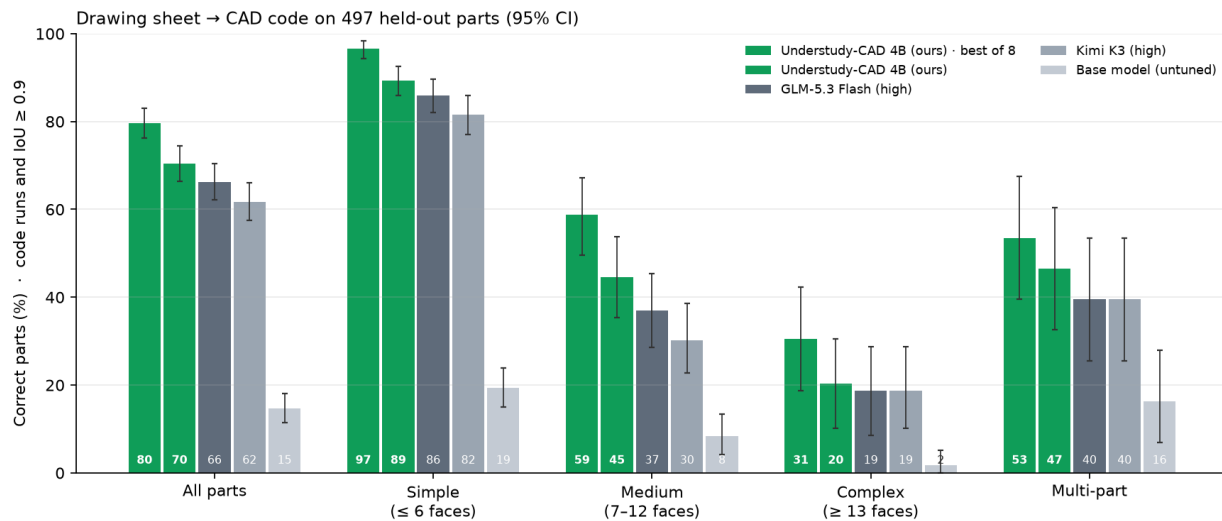


Figure 2: The same numbers by part complexity, with 95% confidence intervals. Our lead is widest exactly where the frontier models are weakest: medium parts (58.8% vs 37.0%) and complex parts (30.5% vs 18.6%).

Two observations. First, a single sample already beats both frontier models while costing $5\times$ less than GLM-5.3 Flash and $194\times$ less than Kimi K3, and it answers in 1.9s against 3.7s and 10.4s. Second, best-of-8 costs about as much per part as Flash and adds 9.3 points, because the verifier recovers 97% of what sampling made available (its oracle, the share of parts where any of the eight was correct, is 81.9%).

The verifier helps any model. Running the frontier models with the same best-of-4 procedure on the 65 medium and complex parts lifts Kimi K3 from 27.7% to 48.3% and GLM-5.3 Flash from 32.3% to 45.9%. It is a general technique; it is simply cheap for a 4B model and expensive for a frontier one (\$251.67 per 1,000 parts for K3).

Learning curve. Figure 3 scores intermediate checkpoints on the same parts. The untuned model gets 13.2%; after 200 steps (3,200 examples, 11% of the epoch) it matches Kimi K3, and it passes GLM-5.3 Flash around step 1,000.

7 Did it overfit, or memorise the benchmark?

Three checks.

Training. One pass over the data; nothing was seen twice. Validation loss fell monotonically to the last step ($0.2119 \rightarrow 0.1583$) and validation token accuracy rose to 94.4%, so the run stopped while still improving rather than after a turn upward.

Exact overlap. Zero benchmark source parts appear in training, and zero training parts share a benchmark part’s geometry signature, by construction.

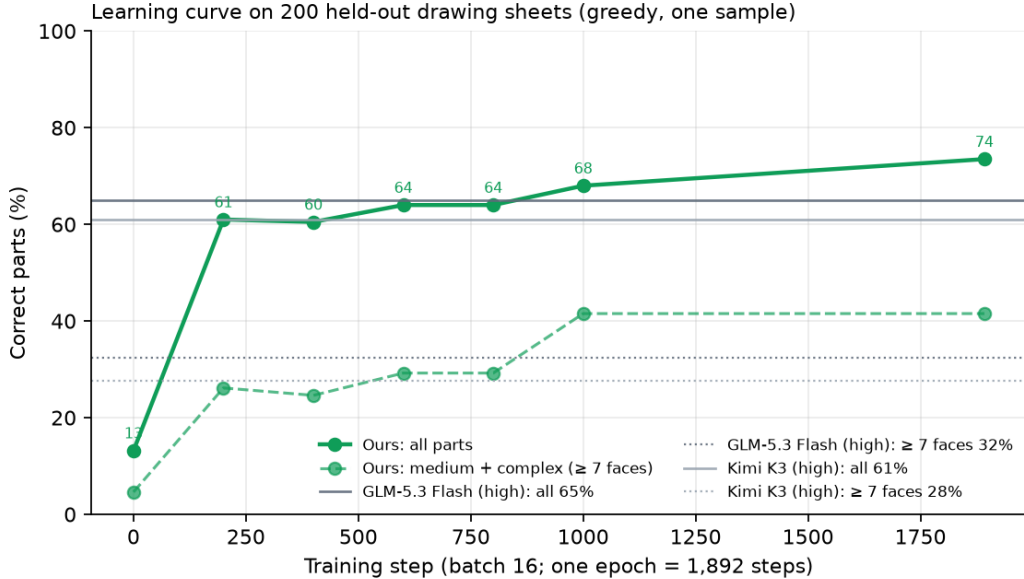


Figure 3: Accuracy against training step, all on the same 200 held-out parts, greedy decoding. Checkpoints were served as four LoRA adapters from a single deployment and benchmarked in parallel.

Near-duplicates. 57 of 500 benchmark parts have a training part within 1% on every bounding-box axis with the same face and part count (none within 0.1%). Those parts are easier for every model, including ones that never saw our data, so the question is whether *our* advantage depends on them. It does not (Table 3): our margin over the best frontier model is +13.5 points on the 443 parts with no near-duplicate, the same as the +13.5 overall.

	Near-duplicate (57)	No near-duplicate (443)
Ours, best of 8	96.5	77.4
Ours, 1 sample	93.0	67.5
GLM-5.3 Flash (high)	84.2	63.9
Kimi K3 (high)	73.7	60.3

Table 3: Correct parts (%) split by whether a near-duplicate exists in training (`scripts/leakage_check.py`).

8 Limitations

The model is trained and tested on one distribution: DeepCAD-style mechanical parts rendered by our own renderer. Nothing here shows it transfers to scanned drawings, other CAD dialects, or drawings with dimension callouts instead of a stated bounding box. The bounding box is given, which removes the global scale problem. Complex parts remain the weak point at 30.5%, so most of the task is still open. Finally, our cost figure assumes a busy GPU; at low volume a per-token API is cheaper than a dedicated one.

9 Reproducing this

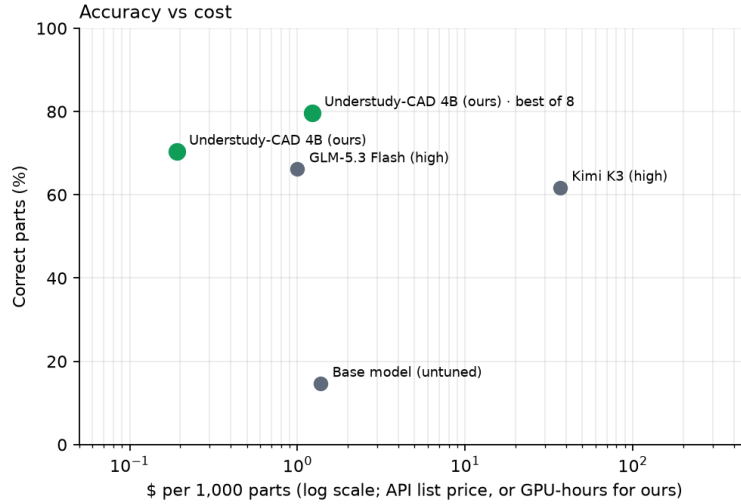


Figure 4: Accuracy against cost per 1,000 parts (log scale). Our two operating points sit above and to the left of every alternative.

```
python scripts/audit_cadcoder.py # run every reference program, check it against its spec
python -m understudy.cad.splits # benchmark/training splits, held out by source part and geometry
python -m understudy.cad.render --split bench # drawing sheets
python -m understudy.cad.build_vlm # training rows + images
cd training/vlm && baseten train push --config config_vlm.py --job-name sft # LoRA SFT on H100(s)
./scripts/deploy_vlm.sh <job_id> H100 # merge -> vLLM on Baseten
python -m understudy.cad.bench --modality image --lanes specialist --n 0 --shots 0 --tag ours-img
python scripts/leaderboard.py --latest sota-img,ours-img --common --out-json data/demo/scoreboard.json
python scripts/plot_results.py && python scripts/plot_curve.py && python scripts/leakage_check.py
```

Every benchmark run in the repository keeps its per-part rows, including the program each model wrote, so any number here can be recomputed or spot-checked.